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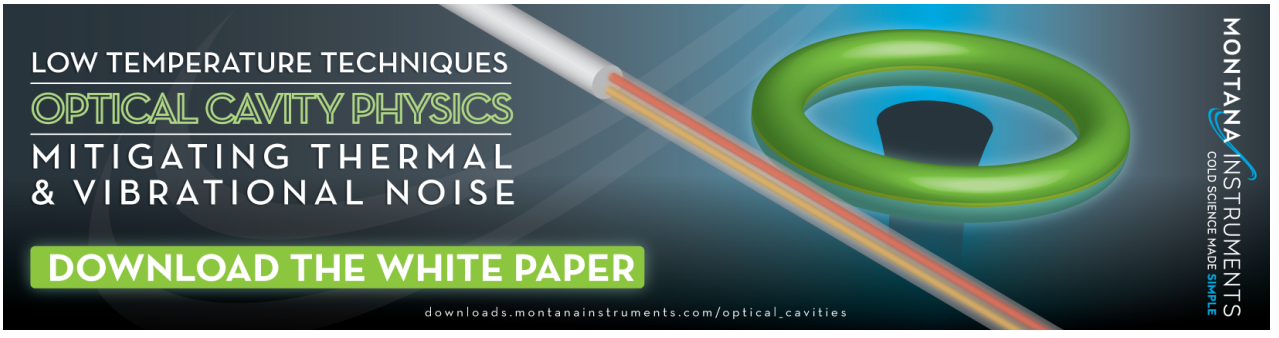
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Spin crossover in iron(II)-containing complex compounds under a pressure (Review Article)

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The spin crossover phenomenon consists in switching between two spin states of the central ion of a transition 3*d*- metal in a molecular complex. The spin crossover can be induced and controlled by temperature, pressure, light radiation etc. The present review addresses all aspects of the effect of hydrostatic pressure on spin-crossover properties of iron(II) complex compounds. The applied hydrostatic pressure allows us to influence the crystal lattice, revealing the role of elastic fields in the cooperative interaction of spin-crossover complexes, and to change directly the volume characteristics of the complex molecules, the distances between the central ion and ligand atoms, and thus affect the strength of the crystal field. The experimental results for all types of iron(II)-containing spin-crossover structures are discussed in details, and the theoretical consideration of the observed effects is given. The physical principles of spin crossover at the molecular level are adequately described by the ligand field theory while the macroscopic properties are understood on the basis of electron–phonon coupling and elastic properties of the crystal lattice. © *2014 AIP Publishing LLC*.
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1. Introduction

Spin crossover (SC) is intrinsic to all transition metal ions with the electron configurations from d^4 to d^7 (and also d^8 if the symmetry of the molecule is below O_h) located in a molecular coordination complex.[1] The SC phenomenon was discovered by Cambi almost 80 years ago.[2–4] He observed the dependence of the electronic states in Fe(III)tris(dithiocarbamate) on temperature, which was determined by the magnitude and behavior of the ligand field. More than 30 years after, it was found that the spin state of bivalent iron in the compound $[Fe(phen)_2(NCS)_2]$(phen = 1,10-phenanthroline) undergoes a change $T_{2g} \leftrightarrow A_{1g}$ (O_h) upon changing temperature.[5] This marked the beginning of intensive study of the spin transition phenomenon. SC has been found in the complexes of divalent and trivalent iron, divalent and trivalent cobalt, nickel, chromium, and even manganese.[6–8] Starting from late 60s, the divalent iron (Fe(II)) became the material of choice for SC studies.[9–23] At first it was due to the availability of the most common research method at that time—Mössbauer spectroscopy, and then due to the possibilities for application of SC in Fe(II) for recording and storage of information at the molecular level. Currently, along with the complexes of divalent iron, the complexes of all 3*d*-ions are widely studied.

To understand the nature of SC within the ligand-field approximation, it is necessary to consider the behavior of Fe(II) in an octahedral field (Fig. 1). In an octahedral field, five orbital levels of Fe(II) split into a triplet t_{2g} and a doublet e_g.[24] The t_{2g} levels are positioned below the e_g levels.[25] Splitting between the doublet and triplet depends on the magnitude of the ligand field, which is determined by the distances between the central ion and the ligands and the parameters of the central ion and the ligands ions.[26–28] For octahedral symmetry of the ligand environment and the iron(II), the crystal field $10Dq$ depends on the metal-ligand distance as $1/r^n$, where $n = 5$–6.[29]

When the energy of electron pairing Π is large compared to $10Dq$, electrons occupy five *d*-orbitals according to Hund's rule of maximum spin multiplicity. This state of the ion is the high-spin (HS) state ${}^5T_{2g}(t_{2g}^4 e_g^2)$ with the spin $S = 2$. If $10Dq$ is stronger than electron-electron repulsion, the six electrons occupy t_{2g}-orbitals. This state will be the diamagnetic low-spin (LS) state ${}^1A_{1g}(t_{2g}^6)$ with $S = 0$.

For each main HS and LS state, the energy of perturbed electrons states can be calculated as a function of r through $10Dq(r)$ from the following equation:

$$10Dq(r) = 10Dq(r_0)(r_0/r)^6, \quad (1)$$

where r_0 is the equilibrium distance corresponding to the ground state. It should be noted that, in contrast to $10Dq$, the electron-electron interaction Π is weakly dependent on r.

The consequence of these considerations is that the adiabatic potentials of the two spin states are shifted horizontally

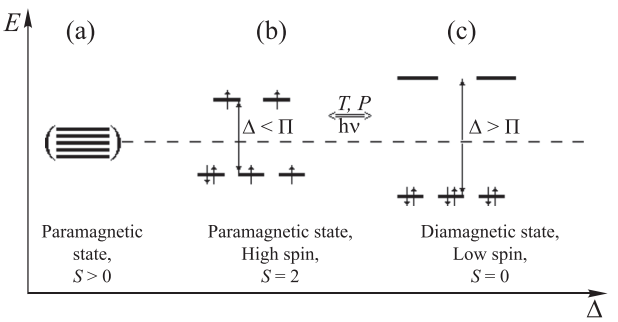


FIG. 1. Splitting scheme for the configuration $(3d)^6$ Fe(II) ion in an octahedral field (Δ is the energy splitting of the ligand levels, Π is the energy electron-electron repulsion): state of a free ion (a), weak field (b), strong field (c).

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and vertically relative to each other. The potential well for the HS state $^5T_{2g}$ is always shifted toward larger metal-ligand distances compared to the LS $^1A_1$ state, irrespective of the ligand type.

The transition of complexes from one spin state to another can be realized by changing the occupancy of the terms $^1A_{1g}$ and $^5T_{2g}$. This can be achieved by changing temperature. At low temperatures, the main state is always $^1A_1$. With increasing temperature, the occupancy of the $^5T_{2g}$ term increases and the transition to the HS state occurs. Thus, this transition is a temperature-induced SC (TISC). At constant temperature, a SC can be also induced by changing pressure. If a sufficiently high pressure is used at a constant temperature, the shift of the adiabatic potential of the HS state with respect to the adiabatic potential of the LS state leads to a phase transition from the HS state to the LS state. Such a transition is called "pressure-induced spin transition" (PISC). In this case, the pressure changes the ligand field by changing r, resulting in a SC. Since the pressure directly changes the crystal field, this method is highly informative for studying phase transitions associated with a change in the spin state of the magnetic ions.

2. Effect of hydrostatic pressure on the "high spin–low spin" phase transition

Applying pressure to explore spin-crossover compounds began in the 60s of the last century. In these experiments, diamond anvils were used to apply a pressure, and electronic or Mössbauer spectroscopy was employed as the measuring method.[30–33] These were the experiments on inducing SC by applying a pressure. TISC under pressure has first been introduced in Ref. 34. The effect of pressure on TISC behaving as a first-order phase transition has first been investigated in Ref. 35. The interaction parameters and pressure-induced volume change for the systems with smooth and sharp transitions have first been considered in the lattice model.[36] Reference 37 has experimentally studied the effect of pressure on SC by accounting the influence of lattice interactions and changes in the ligand field and polarization effects upon changing pressure. References 38 and 39 addressed the effect of pressure on a two-step spin transition. Relaxation phenomena under pressure have been studied in Ref. 40 for the case of dilute systems $[Fe_{0.1}Zn_{0.9}(ptz)_6](BF_4)_2$. It has been was found that the relaxation rate increases by an order of magnitude upon a pressure increase of 0.1 GPa. The X-ray studies that have been carried out on the compound in Ref. 41 have revealed a structural transition at a pressure above 0.025 GPa even in the most diluted compound, thereby inducing the spin crossover.

In Refs. 42 and 43, the effect of pressure on the spin state of several spin-crossover compounds has been studied using the X-ray absorption method (XANES) at room temperature. The pressure dependence of the TISC temperature for different compounds has been also analyzed there.

Despite of the above-mentioned studies examining the effect of pressure on the spin state of magnetic ions in coordination compounds, a number of questions have remained unanswered at the beginning of this century. These were the questions regarding the experimentally obtained behavior of the spin-crossover compounds under pressure, the theoretical interpretation of the effect of pressure on the TISC, and the regularities in the pressure-induced spin crossover at constant temperature.

Development and design of high hydrostatic pressure chambers for magnetic measurements using first a vibrating magnetometer[44] and then a SQUID magnetometer significantly broadened and deepened the scope of pressure studies, allowing to establish a general picture of the pressure effect on TISC.

By the time the studies assessed here have begun, there was an uncertain situation in the interpretation of the effect of pressure on TISC and PISC. Generally, it was believed that the pressure always increases the temperature of TISC. This was justified by the fact that the distance between the central ion and ligand environment in the LS state is shorter than that in the HS state and the volume of the LS phase is less than that of the HS phase. It was also believed that pressure always reduces the width of the hysteresis up until the critical temperature is reached, at which the hysteresis collapses and TISC becomes a second order phase transition. For the pressure-induced spin transition, it was believed that the pressure always stabilizes the LS state. Nevertheless, there were a couple of experimental studies that contradicted these ideas. These were the study by König *et al.*,[35] where an increase of hysteresis under pressure has been observed, and the study by Drickamer and Slichter,[45] where at room temperature under pressure first the LS phase was induced, and then a slight increase in the fraction of the HS phase was observed. Both in the first and in the second case it has been assumed that the anomalous behavior of the hysteresis is due to non-hydrostaticity of the pressure created by anvils. Therefore there was an urgent need for a comprehensive study of all types of transitions under pressure, using hydrostatic pressure and specific structures of the compounds under study.

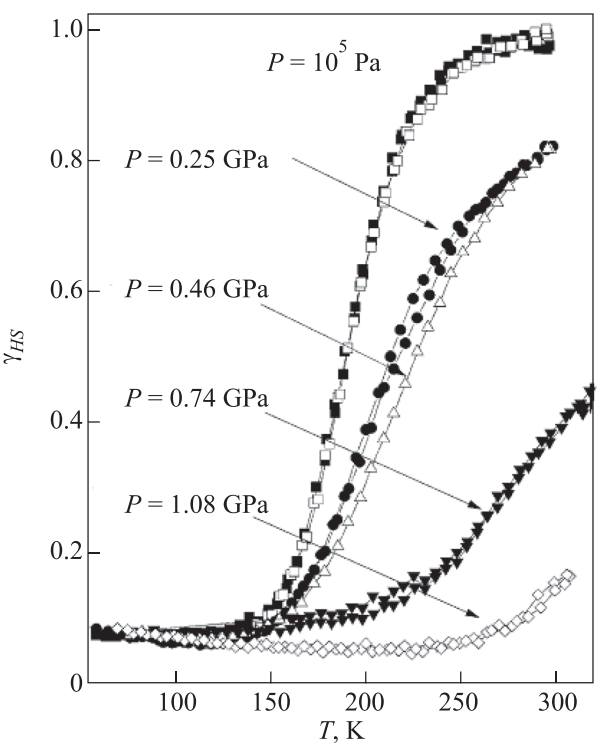


FIG. 2. Fraction of the high-spin phase (γ_{HS}) as a function of temperature for $Fe(PM\text{-}A_2A)_2(NCS)_2$ at different pressures.

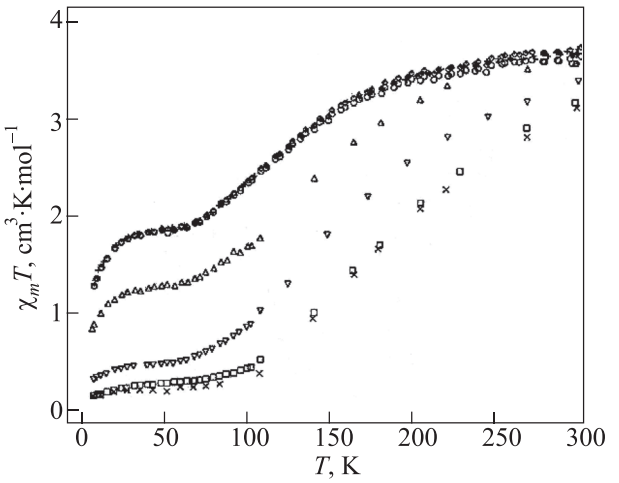


FIG. 3. Temperature dependence of the product $\chi_m T$ for Fe(PM-A)$_2$(NCS)$_2$ at different pressures, GPa: 0.1 (○); 0,16 (△); 0.6 (▽); 1 (□).

2.1. Effect of pressure on TISC

Mononuclear spin-crossover compounds under pressure. Most of the spin-crossover compounds contain mononuclear complexes interconnected by non-covalent bonds. These compounds exhibit a huge range of spin transitions, which are characterized by different widths of the transitions and hysteresis.[1,46] To elucidate the effect of pressure on the TISC in mononuclear compounds, a number of purposeful studies has been carried out.[12,13,16,35,36,47–52] The main conclusion of these studies can be exemplified by the studies of spin transitions in FeL$_2$(NCS)$_2$ compounds with very flexible ligand molecules, which exhibit a broad range of various behavior—from HS state at all temperatures to the presence of temperature-induced transitions with different hysteresis widths.[12] The behavior of TISC in these compounds under pressure is shown in Figs. 2–5.

As can be seen in these figures, in spite of the isostructural nature of these compounds, they exhibit a variety of different temperature dependences of the magnetic susceptibility at atmospheric pressure, from an incomplete spin transition in Fe(PM-A)$_2$(NCS)$_2$ to a spin transition with a huge hysteresis in Fe(PM-PEA)$_2$(NCS)$_2$. We can also see a different behavior of TISC under pressure for these compounds. In the absence

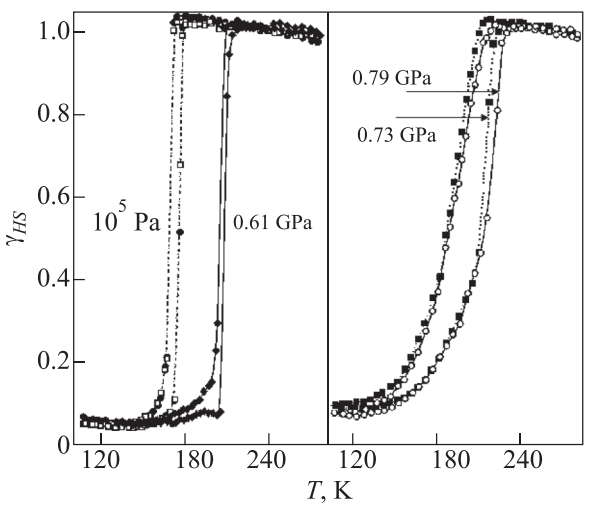


FIG. 4. Temperature dependence of the fraction of the high-spin phase (γ_{HS}) for Fe(PM-BiA)$_2$(NCS)$_2$ at different pressures.[12]

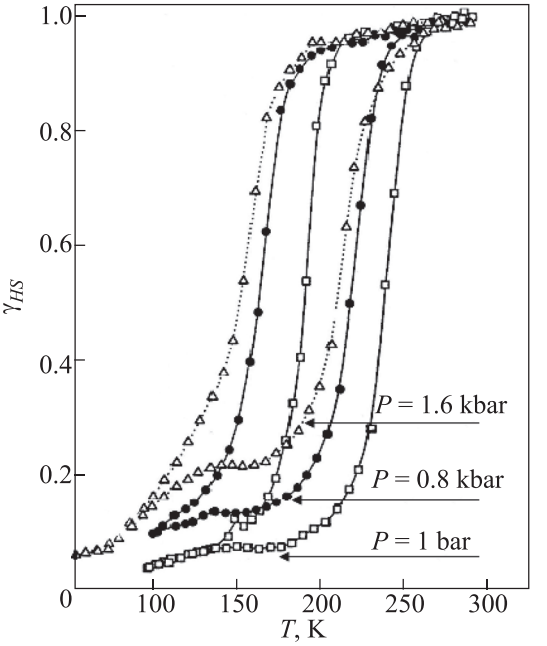


FIG. 5. Temperature dependence of the fraction of the high-spin phase (γ_{HS}) for the compound Fe(PM-REA)$_2$(NCS)$_2$ at different pressures.[12]

of hysteresis, the temperature of TISC increases and the transition broadens in temperature with increasing pressure. Such a behavior of hysteresis-free transitions has been also observed in [CrI$_2$(depe)$_2$][52] and Fe[H$_2$B(pz)$_2$]$_2$(bipy).[20] It corresponds to the generally accepted picture of the LS phase being stabilized by pressure and the slope of $\gamma(T)$ being lowered upon increasing pressure.

A transition which is incomplete at atmospheric pressure (Fig. 3) becomes more and more complete under pressure up to the appearance of a complete transition. This is accompanied by a continuous increase of the transition temperature. Such behavior of incomplete transitions under pressure has been also confirmed on other compounds: [Fe(mtz)$_6$](BF$_4$)$_2$,[53] [Fe(abpt)$_2$(NCS))$_2$] polymorph B[50,51] and {Fe^{II}(L)$_x$[Ag(CN)$_2$]$_2$}·G, where L = 4.4′-py and bp and G = pz.[54]

In the presence of a hysteresis, the behavior of TISC can vary, as shown in Figs. 4 and 5. In the first case, for the compound Fe(PM-BiA)$_2$(NCS)$_2$ (Fig. 4), the hysteresis is first

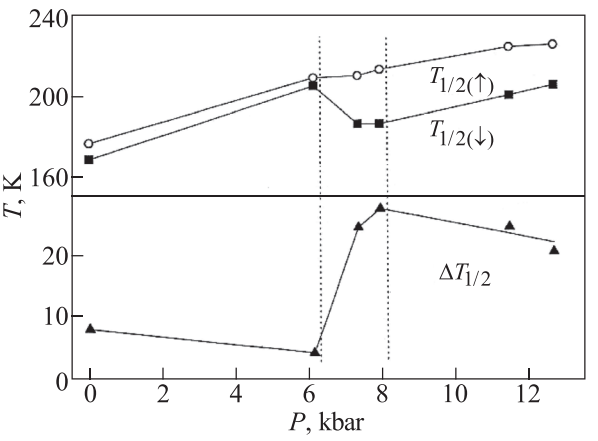


FIG. 6. Pressure dependences of the TISC transition temperature in Fe(PM-BiA)$_2$(NCS)$_2$ upon increasing ($T_{1/2}\uparrow$) and decreasing ($T_{1/2}\downarrow$) temperature (top) and the hysteresis width (bottom).[12]

reduced, then there is a transition region where the hysteresis increases, and then it decreases again (Fig. 6).

The transition region can be due to various reasons, one of which is structural changes. However, in any case, it is indicative that the transition temperature $T_{1/2} = (T_{1/2\uparrow} + T_{1/2\downarrow})/2$ decreases with increasing the hysteresis. A decrease in hysteresis with increasing pressure has been observed in a number of experiments, for example, for $Fe(phen)_2(NCS)_2$,[49,53] as well as for more complex, one-, two-, and three-dimensional compounds, which will be discussed later in the text. However, after its disappearance, the hysteresis always reappears.

In the second case (Fig. 5), the hysteresis increases with increasing pressure, while the transition temperature decreases. The drop in the transition temperature with increasing pressure has also been observed in the three-dimensional compound $Fe(pmd)[Au(CN)]\cdot 2H_2O$, which will be discussed later in the text.

With regard to an increase in hysteresis under pressure, it has also been observed in $Fe(phy)_2(BF_4)_2$, however, at the same time, the transition temperature increased with increasing pressure.[13] An increase in the hysteresis with increasing $T_{1/2}$ has also been observed in $[Fe(bt)_2(NCS)_2]$.[55]

Compounds in which the HS state is observed at all temperatures, but a spin transition is induced under pressure

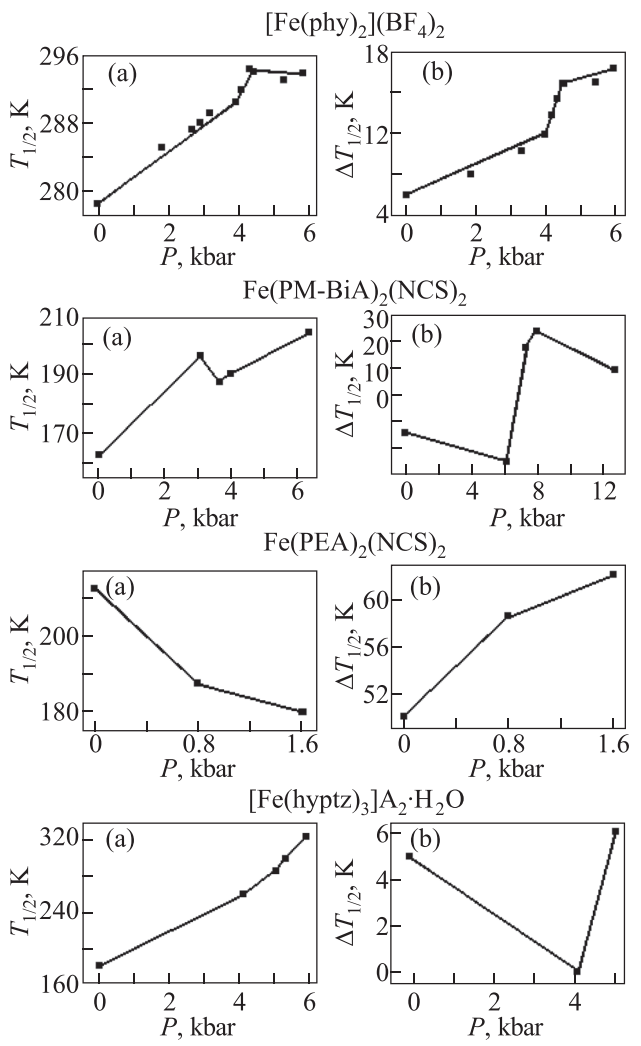


FIG. 7. Pressure dependences of the crossover temperature $T_{1/2}$ and the hysteresis widths $(\Delta)T_{1/2}$ for the compounds $[Fe(phy)_2](BF_4)_2$, $[Fe(PM\text{-}BiA)_2(NCS)_2]$, $[Fe(PEA)_2(NCS)_2]$, $[Fe(hyptz)_3]A_2\cdot H_2O$.

hold a special place. In these compounds, first, a partial transition under pressure usually occurs, and then as pressure increases, the transition becomes complete and shifts upward in temperature.[50,51]

In all the above cases, the LS state is stabilized under pressure, which is consistent with the general concept that the volume of a compound is reduced under pressure and a coordination complex in the LS state has a lesser volume. However the change in the crossover temperature and hysteresis width is different for different mononuclear compounds. The above behavior of SC can be summarized with the data in Fig. 7, which show various patterns of the behavior of crossover temperature and hysteresis under pressure.[16] However, the effect of pressure on spin-crossover compounds goes beyond this set of patterns. For instance, the study of the effect of pressure on the two-dimensional polymeric compound $[Fe(btr)_2(NCS)_2]\cdot H_2O$ has revealed the phenomenon of pressure-induced HS state in the low-spin region, up until the disappearance of the LS state.[14] And when, in the above-mentioned Ref. 55, the effect of pressure on the compound $[Fe(bt)_2(NCS)_2]$ was studied, the appearance of an asymmetric hysteresis at high pressures has been observed, where the transition is sharp upon increasing temperature and broadened when the temperature is decreasing.

Besides mononuclear compounds, the pressure effect was also investigated in binuclear compounds and compounds of various dimensions.[56–58]

Study of the effect of high pressure on binuclear compounds revealed that the intermolecular interactions between HS and LS molecules lead to the formation of domains, stabilizing the crystal lattice.[38,59,60]

Along with the use of static pressure, the studies of the effect of impulsive pressure on the spin state of the coordination complexes, for instance, $[Fe(phen)_2(NCS)_2]$,[61] have been also reported in literature. In Ref. 61, it has been shown that a relatively low impulsive pressure of 0.02 GPa is capable of inducing an almost complete spin transition on hysteresis branches. Such an effect can be applied to switch the spin state in the devices employing spin-crossover (SCO) materials.

In order to obtain materials suitable for possible applications in information storage or color imaging devices, one-, two-, and three-dimensional spin-crossover compounds have been synthesized. Study of the effect of pressure on TISC in these compounds sheds light on the role of intra- and interchain, interplanar, and bulk interactions in the characteristics of TISC.

1*D spin-crossover polymeric compounds under pressure*. One of the first studies of the effect of pressure on one-dimensional compounds was carried out on [Fe(hyetrz)$_3$](4-chlorophenylsulfonate).[62] This compound has a partial spin transition with a hysteresis of 10 K at $T_{1/2} = 100$ K. Under pressure, the transition is shifted upward in temperature while the hysteresis first decreases and then increases with increasing pressure. The steepness of the transition remains practically unchanged. A change in the transition temperature is sufficiently large and is 230 K/GPa. A similar behavior has been observed for the one-dimensional compound $[Fe(hyptrz)_3]$(4-chlorophenylsulfonate)$_2\cdot H_2O$, where hyptrz = 4–(3′-hydroxypropyl)–1,2,4-triazole.[63]

The behavior of TISC under pressure, which is different from the above-described cases, has been observed in

$[Fe(hyptrz)_3](4\text{-chloro-3-nitrophenylsulfonate})_2\cdot 2H_2O$, which exhibits a 50% transition with a width of 48 K at $T_{1/2} = 144$ K at atmospheric pressure.[64] Despite the fact that the transition is stable (i.e., reproduces upon repeated temperature cycling), it changes shape, becoming broadened with a greatly reduced hysteresis, when placed in silicone oil. Under pressure, the hysteresis is restored, and the transition temperature is shifted up at a tremendous rate of 1000 K/GPa. Such a behavior of this compound under pressure is still not understood and requires further study.

In order to provide a complete picture, the behavior of 1D compounds with an incomplete transition or no transition was investigated on $[Fe(4,4'\text{-bipy})(bt)(NCS)_2]$ (incomplete transition) and $[Fe(atpy)(bt)(NCS)_2]\cdot bt$ (paramagnetic compound over the entire temperature range).[65] The behavior of these compounds under pressure is similar to that of mononuclear compounds. Specifically, for an incomplete transition, the step is reduced under pressure, then the transition becomes a 50% transition and shifts up in temperature. For the paramagnetic compounds, upon applying a pressure, first a step appears, which then behaves as in the previous scenario.

Evidently, the behavior of TISC in 1D materials is determined by the interaction within and between the chains. Moreover interchain interactions are largely determined by hydrogen bonds. This fact is confirmed by either an increase in the steepness of the transition or an increase in the hysteresis with increasing the water content in the above compounds. In many cases, the loss of water leads to the disappearance of TISC. The fact of the influence of intra- and inter-chain interactions on the behavior of TISC in 1D compounds has been considered in the microscopic theory developed by Klokishner *et al.* in Ref. 66. According to this model, the magnitude and the competition of near- and long-range interactions determine the temperature, slope, and hysteresis of the transitions. This model successfully describes the main features of the effect of pressure on TISC in 1D systems.

2D spin-crossover polymeric compound under pressure. One of the prominent examples of 2D compounds is $[Fe(btr)_2(NCS)_2]\cdot H_2O$, which has a sharp TISC with a large hysteresis at atmospheric pressure. This crossover is presumably associated with a structural transition. We say "presumably" since the sample cracks into small fragments upon the transition and determination of the structure after the transition becomes impossible. The effect of pressure on TISC in this compound is shown in Fig. 8.[14] The figure shows that with increasing pressure, the transition shifts upward in temperature and becomes less sharp. Then, in the region of the LS state, the high-spin state appears, the fraction of which grows with increasing pressure. At a pressure of 1.05 GPa, the transition with hysteresis is not observed, but the curve $\chi_m T(T)$ gradually decreases with decreasing temperature, which can be interpreted as a smooth, strongly stretched spin transition or as a reduction in the susceptibility with decreasing temperature under the influence of antiferromagnetic interactions. After removal of the pressure, the transition with a hysteresis appears; it is incomplete, although eventually relaxes to the LS state. After 175 h incubation of the sample at room temperature, the residual ratio of the HS phase was 30%.

Such a behavior of TISC under a pressure is rather unexpected since the pressure must stabilize the LS state. The

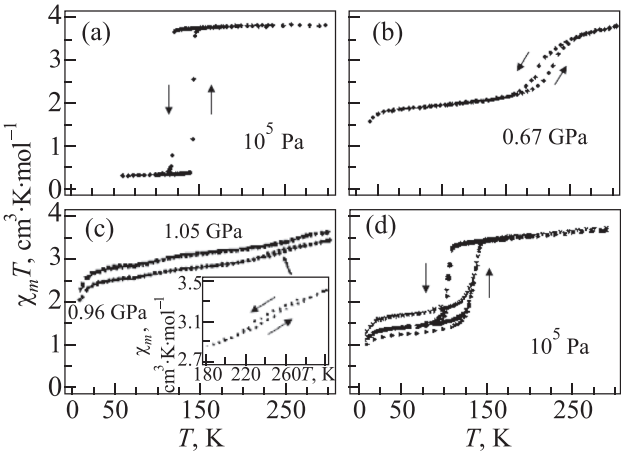


FIG. 8. Temperature dependences of $\chi_m T$ for $[Fe(btr)_2(NCS)_2]\cdot H_2O$[89] at different pressures.

observed effect may be caused by non-isotropic compressibility of the octahedron surrounding the Fe(II) ion. This possibility has been considered by Kambara[67] and Shelest *et al.*[68,69]

Of a particular interest are the studies of a group of 2D bimetallic compounds $\{Fe(3\text{-}Xpy)_2[M^{II}(CN)_4]\}$ (py denotes pyridine; X = F, Cl, Br or I; M^{II} = Ni, Pd, Pt) under a pressure, which exhibit sharp TISC with a wide hysteresis at sufficiently high temperatures.[70,71] Compounds of this group with X = F have a transition temperature above 200 K at atmospheric pressure. Under pressure, as follows from the data shown in Fig. 9, [70] the spin crossovers for all the compounds are shifted to the right and the transition temperatures become higher than room temperature at a pressure of about 0.5 GPa. At the same time, the slope of the transitions remains practically unchanged, which is very important for practical purposes.

The effect of pressure on TISC without hysteresis in 2D compounds has been studied using $[Fe(brb)_2(NCS)_2]\cdot 0.5MeOH$ as an example and has been found to cause an increase of the transition temperature.

As follows from the above description of the effect of pressure on the 2D compounds, we can conclude that it agrees with the general idea of the behavior of TISC under a pressure. The observed appearance of the HS phase in the region of the low-spin phase in $[Fe(btr)_2(NCS)_2]\cdot H_2O$ under pressure is most probably determined by changing the symmetry of the coordination environment of Fe(II) ion, rather than the dimensionality of the whole system.

3D spin-crossover polymeric compounds under pressure. To date, there are two sets of measurements of TISC behavior in 3D spin-crossover compounds under pressure. One of them is related to the studies of the effect of pressure on the compounds $[Fe(L)_xAg(CN)_2]\cdot G$, where L = 4,4′-py, $x = 2$; L = bp, $x = 2$; L = bpac, $x = 2$; and L = pz, $x = 1$; G = pz. The latter compound, $\{Fe^{II}(pz)[Ag(CN)_2]_2\}pz$ is diamagnetic up to a temperature above room temperature, so the first two compounds were investigated.[54] The structure of these compounds consists of interpenetrating cells, which contain parallel stacks of $\{Fe[Ag(CN)_2]_2\}$ layers, which are aligned along different crystallographic directions for the different compounds. This, most probably, determines their different magnetic properties. While the latter compound is

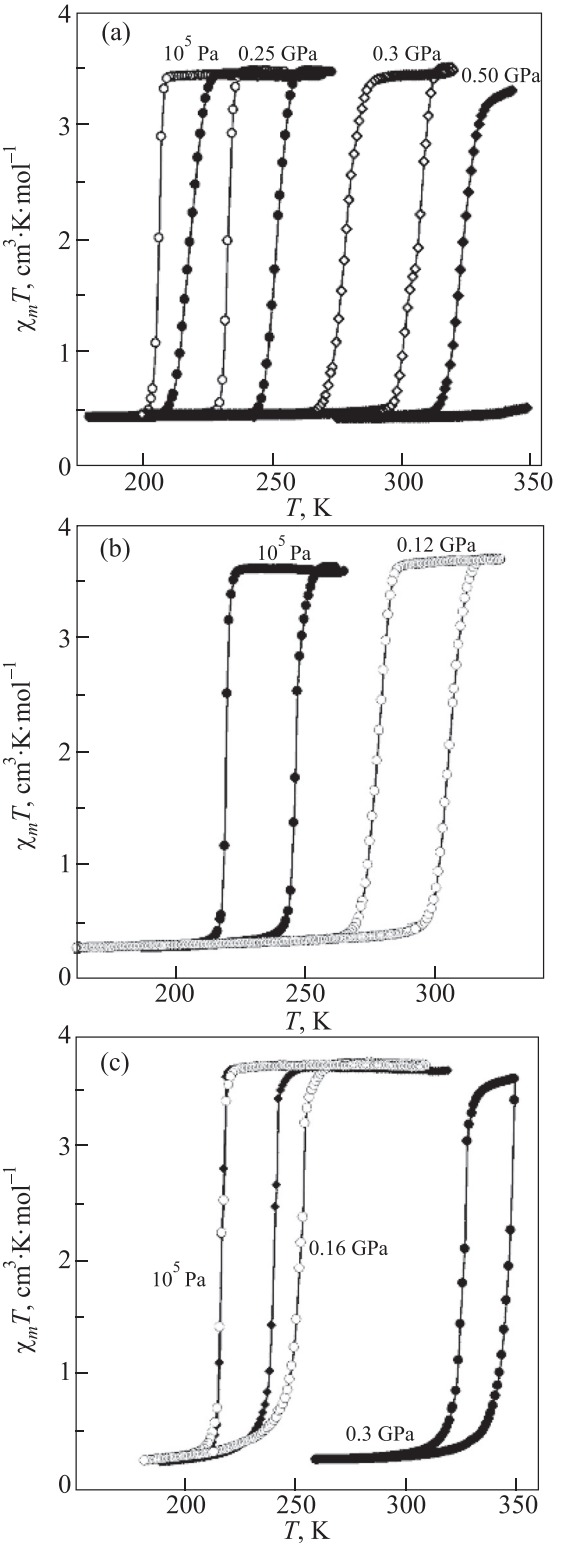


FIG. 9. Temperature dependences of $\chi_m T$ at different pressures: FNi (a), FPd (b), and FPt (c).

diamagnetic, $\{Fe(4,4'\text{-py})_2[Ag(CN)_2]\}_2$ exhibits a 50% hysteresis-free TISC, and $\{Fe^{II}(4,4'\text{-bp})_2[Ag(CN)_2]_2\}$ has a 50% TISC with a huge, 90 K, hysteresis at atmospheric pressure. Moreover, this transition is unstable with respect to temperature cycling. After undergoing spin crossover, a single crystal cracks, turning into a powder. This leads to such changes of the transition, that after a few cycles the transition disappears. The dependence on temperature cycling exhibits an incomplete TISC also in $Fe(bpac)_2[Ag(CN)_2]_2$.[72]

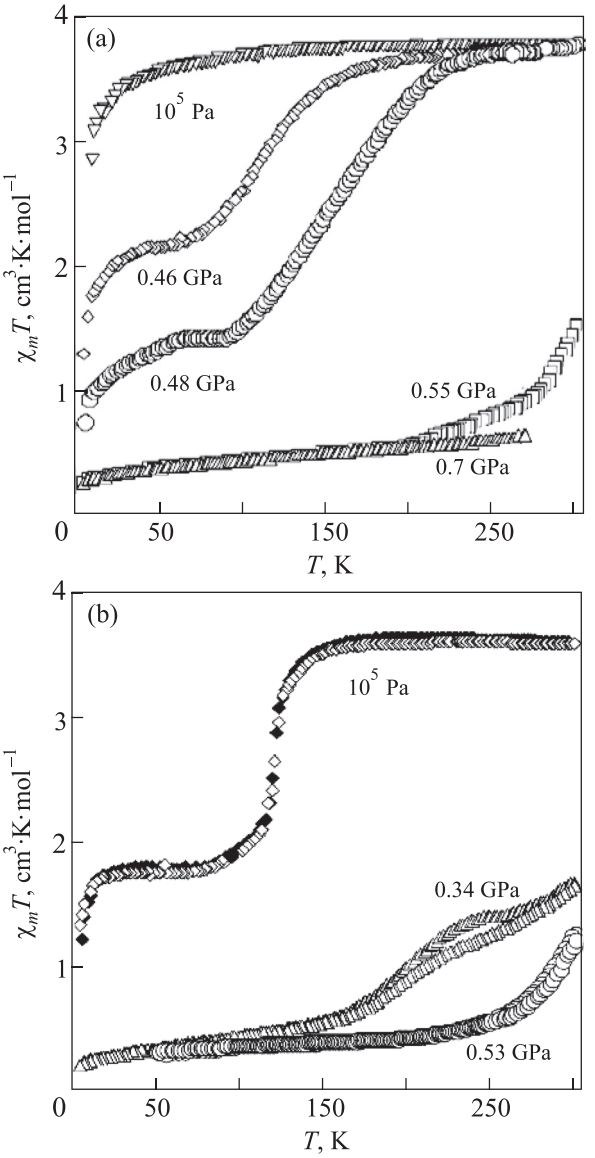


FIG. 10. Temperature dependences of $\chi_m T$: for the powder compound $\{Fe^{II}(py)_2[Ag(CN)_2]_2\}$ (a); for $\{Fe^{II}(4,4'\text{-bp})_2[Ag(CN)_2]_2\}$[54] (b).

The behavior of a powdered compound, which does not exhibit spin crossover, and $\{Fe^{II}(4,4'\text{-py})_2[Ag(CN)_2]_2\}$ have been studied under pressure. Their observed behavior is shown in Fig. 10. It can be seen that both compounds exhibit a similar TISC behavior under pressure, specifically, with increasing the pressure, the amount of the HS phase decreases, and the transition shifts to the right. Even at a low pressure (~0.7 GPa), both compounds become diamagnetic at room temperature.

A more interesting effect of pressure on TISC has been observed in the second group of 3D compounds $\{Fe(pmd)H_2O[M(CN)_2]_2\}\cdot H_2O$, M = Ag, Au.[73] With almost the same TISC at atmospheric pressure and only slightly different transition temperature and hysteresis, the two compounds with M = Ag or Au behave differently under pressure. The TISC behavior under pressure in these compounds is shown in Fig. 11.

In the case of the compound with silver, the transition remains sharp at all pressures, while the transition temperature and hysteresis increase with increasing pressure. Moreover, even at 0.3 GPa, the mid-point of the transition is

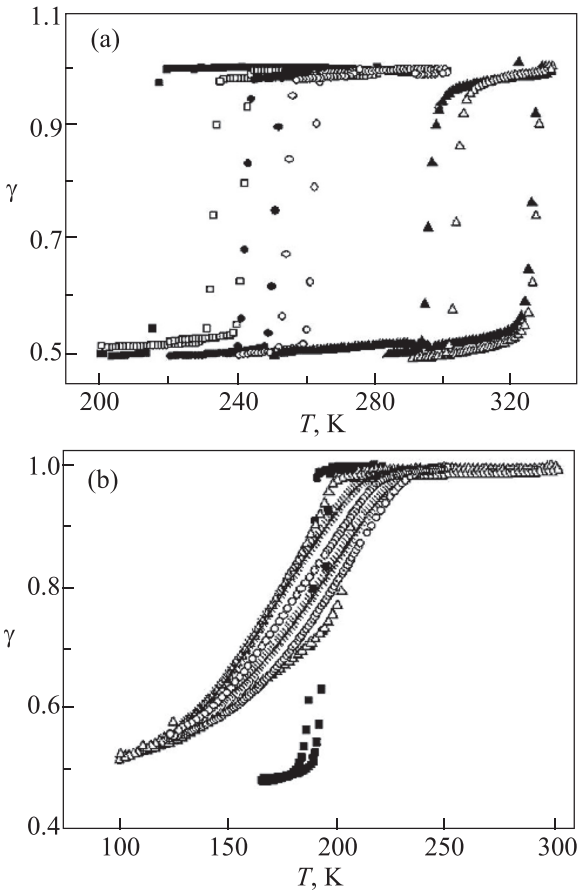


FIG. 11. Temperature dependences of γ for Fe–Ag (a) and Fe–Au (b) at different pressures, GPa: (a)—10^{-3} (■); 0.19 (□); 0.23 (●); 0.27 (○); 0.305 (▲); 0.355 (△); (b)—0.31 (■); 0.44 (△); 0.74 (×); 0.91 (○).[73]

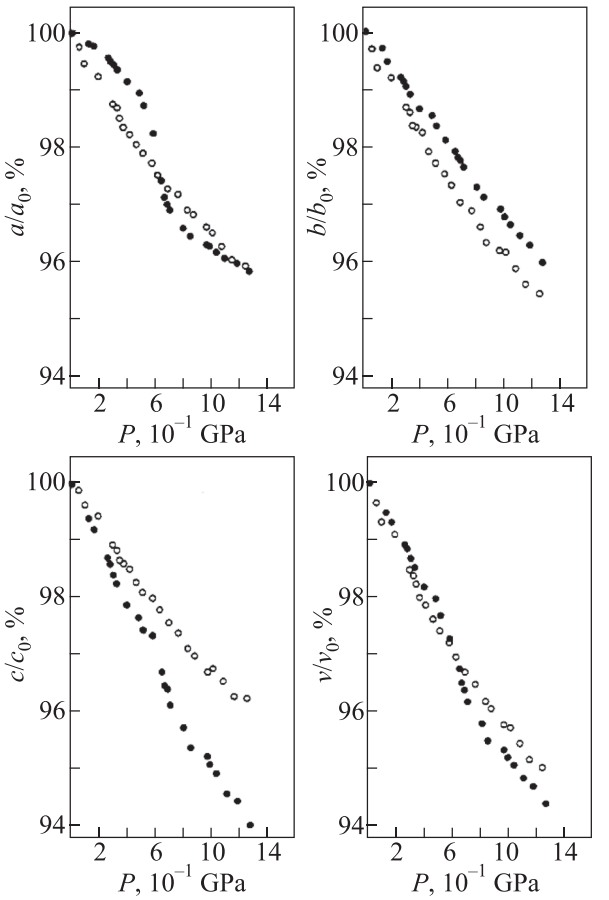


FIG. 12. Pressure dependence of the lattice parameters at room temperature: $Fe(phen)_2(NCS)_2$ (●), $Fe(btz)_2(NCS)_2$ (○).[74]

at room temperature, and the hysteresis width reaches 30 K. A quite different behavior has been observed in the second compound. The transition is stretched in temperature, the hysteresis loop becomes asymmetric, and the transition temperature first increases and then decreases. Apparently, the observed behavior of the two isostructural compounds under pressure is the result of unstable structures and the differences of the interactions parameters associated with it. This will be discussed later in the text, in the section devoted to the theoretical consideration of the effect of pressure on TISC.

2.2. Pressure-induced spin transitions

Along with the study of the effect of pressure on TISC, experimental studies on pressure-induced SC at constant temperature have also been conducted.[30–33] During the early development of this field, the major work was conducted by Drickamer and colleagues.[30,45] The appearance of the LS state has been observed in high-spin compounds under pressure.

The development of this field further continued in Ref. 74, which presents the data on the behavior of lattice parameters under a pressure in two compounds, $Fe(phen)_2(NCS)_2$ and $Fe(btz)_2(NCS)_2$. Comparative data on the behavior of the lattice parameters for these two compounds is shown in Fig. 12. The figure shows that both the lattice parameters and the unit cell volume decrease with increasing pressure. However, while in $Fe(btz)_2(NCS)_2$ both the lattice parameters and the cell volume decrease nearly linearly, in $Fe(phen)_2(NCS)_2$ the parameter a exhibits a jump in the pressure range from 0.5 GPa to 0.7 GPa. This jump produces a change in the cell volume $(\Delta V)_P = 72$ Å^3.

First complete spin crossovers which were induced by pressure at room temperature has been obtained in Ref. 42 using the method of X-ray absorption (XANES) and anvils to create a pressurization. Pressure could only be increased in these experiments, so it is not possible to conclude on the type of transition and the presence of hysteresis in these measurements.

The studies of PISC in which the direction of pressure changes could be varied have been conducted in Refs. 19,23,52,57,72, and 75–78. The first measurements have been carried out on the compound $\{Fe(pmd)(H_2O)[Ag(CN)_2]_2\}\cdot H_2O$,[75] in which a complete transition from the HS state to the LS state and back has been observed. More detailed measurements have been conducted on the compound $Fe(phen)_2(NCS)_2$.[23,76] The measurements in this case were carried out during both the increase of pressure and pressure relief.

Transition curves have been compared with the measurements of the effect of pressure on TISC. Fig. 13 presents the data on the effect of pressure on the volume of unit cell in $Fe(phen)_2(NCS)_2$[74] and the fraction of the HS phase obtained from different measurements of the pressure effect on the TISC.[48,79] The same figure also shows the data on the fraction of the HS phase obtained from the pressure-induced transitions.[23] Since the pressure measurement accuracy is

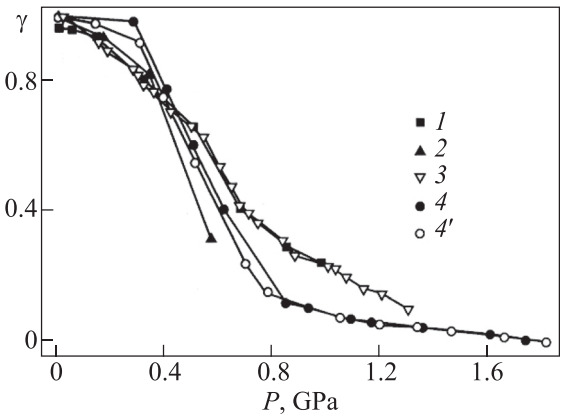


FIG. 13. Fraction of the HS phase as a function of pressure for $Fe(phen)_2(NCS)_2$, calculated using data from different experiments: 1—data from Ref. 79, 2—data from Ref. 48, 3—X-ray structural data from Ref. 74, 4 and 4′—Ref. 23.

±0.05 GPa, a good qualitative agreement of the results of different experiments should be noted.

A different pattern is observed when comparing the results of two types of measurements —behavior of TISC under pressure and pressure-induced SC at room temperature, for $\{Fe(pmd)(H_2O)[Ag(CN)_2]_2\}\cdot H_2O$.[19,75] Fig. 14(a) shows

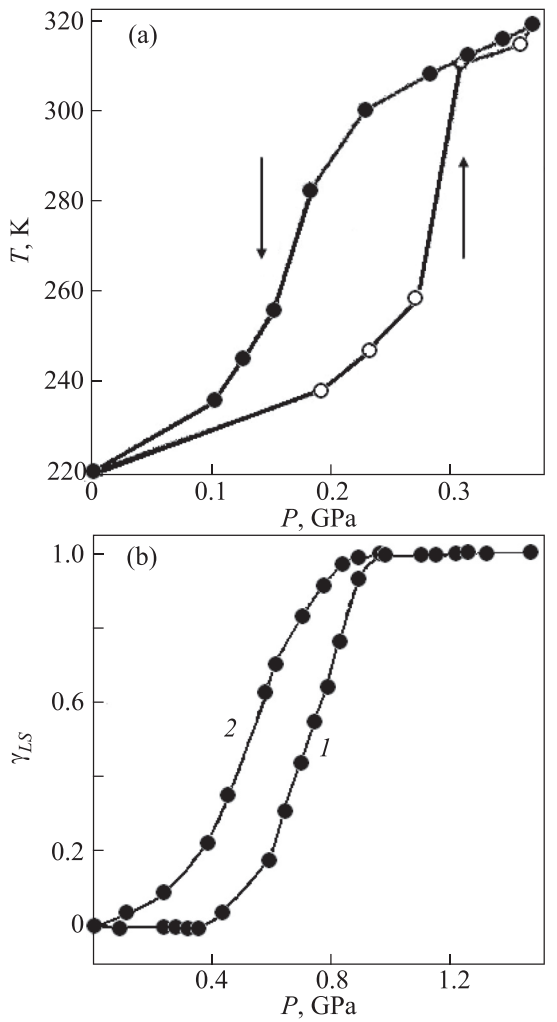


FIG. 14. *T*–*P* phase diagram obtained from measurements of the effect of pressure on TISC in $\{Fe(pmd)(H_2O)[Ag(CN)_2]_2\}\cdot H_2O$ upon increasing and decreasing pressure (a), and the dependence of the fraction of the low-spin phase $\gamma_{LS}(T)$ obtained from the pressure-induced HS–LS transition (b).

the pressure dependences of TISC temperature obtained by increasing and decreasing the pressure, and Fig. 14(b) shows the hysteresis loop for the pressure-induced crossover at room temperature. It can be seen that the former measurements give the crossover pressure of ≈0.25 GPa at room temperature, while the latter produced ≈0.6 GPa. Thus, there is a mismatch in the crossover pressures obtained from different measurements. Such a different behavior of the crossover pressures obtained from the two types of experiments can be due to a number of reasons and requires further detailed studies. These studies began by addressing the behavior of the crossover pressures in the group of isomorphic 2D compounds $Fe(3Fpy)_2M(CN)_4$, where M = Ni, Pd and Pt.[80] These compounds differ from each other by the content of different metal ions, which do not undergo a spin transition. They played a role of creating a different "chemical" pressure in this class of compounds. Indeed, the TISC temperature increased with the ionic radius of the metal. The effect of pressure on TISC for these compounds is shown in Fig. 9. It can be seen that for all the compounds the SC temperature increases with increasing pressure. The pressure-induced crossovers for these compounds are shown in Fig. 15.

Comparison of the crossover pressures at room temperature (according to Figs. 9 and 15) shows a qualitative agreement for the two compounds containing Ni and Pd, and a large difference for the compound with Pt. This experiment indicates that even for isomorphous compounds an unambiguous agreement of the two types of experiments cannot be obtained. According to the current concept of phase transitions, the transition should occur at the same temperature and pressure regardless on which independent parameter is varied to induce the transition. Here, it is not always the case. Therefore, it must be assumed that the parameters *T* and *P* are not always independent.

3. Theoretical description of the effect of pressure on the "high spin-low spin" SC

To date, the effect of pressure on SC has been described within the framework of different approaches and approximations. One of the fundamental approaches is the thermodynamic approach, from which the theoretical analysis of the effect of pressure on SC has begun.[45]

The second successful approach is based on the Ising model and is preferably applicable to 1D systems.[66] The third approach is microscopic, which has been developed by Kambara *et al.*[67,81] and Levchenko *et al.*[69,82,83] Each of these approaches has its advantages and disadvantages, however if we consider the SC as a phase transition, the thermodynamic approach is more productive and better reflects the physical nature of this phenomenon.

Macroscopic models of the SC under pressure. Some of the first studies on the thermodynamic description of the SC behavior under pressure were conducted by Slichter and Drickamer,[45] and Das and Ghosh.[84] The former has been devoted to the description of a pressure-induced crossover, while the latter addressed the behavior of the temperature-induced crossover under pressure based on the Landau theory of phase transitions. However, application of the expressions obtained in this work turned out to be impossible in practice due to the fact that the behavior of an SC in this model is

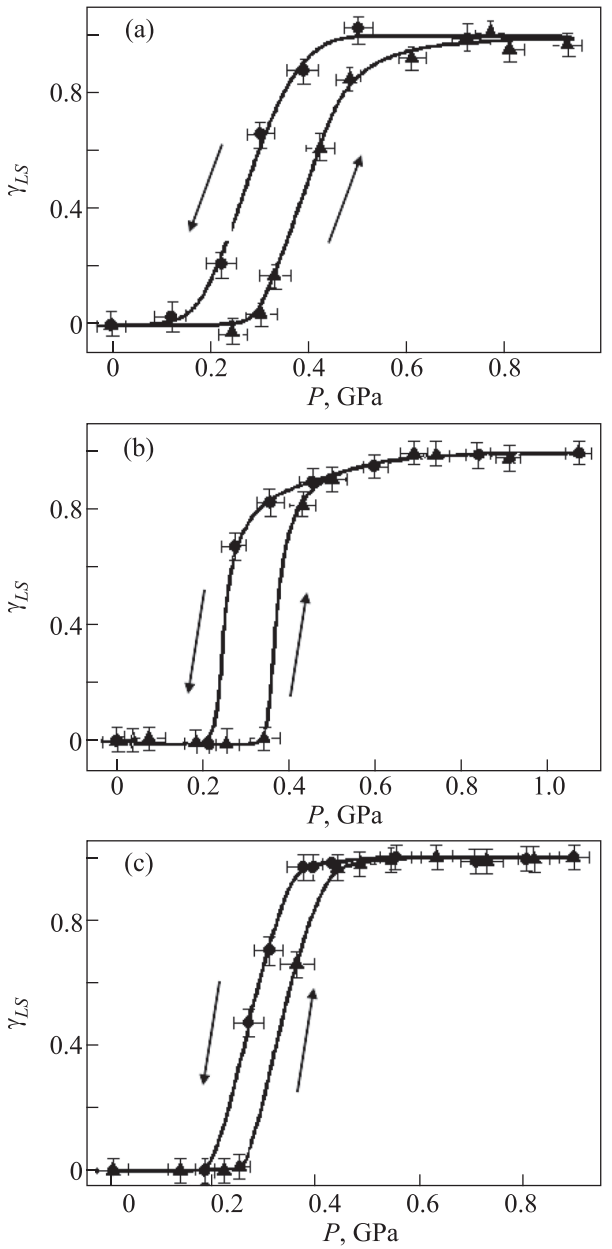


FIG. 15. Pressure dependence of the fraction of the low-spin phase in: $Fe(3\text{-}Fpy)_2Ni(CN)_4$ (a); $Fe(3\text{-}Fpy)_2Pd(CN)_4$ (b); $Fe(3\text{-}Fpy)_2Pt(CN)_4$ (c).

determined by the nonlinear terms of the expansion in the interaction between the elastic strain and the order parameter. This makes it impossible to deduce the direct relation between the pressure and the order parameter. Therefore, in this review we will address only the former approach and present its development in the successive studies.

The thermodynamic approach is based on the statistical probabilities of states. An ensemble of N molecules contains γN of HS molecules and $(1 - \gamma)N$ of LS molecules.

In the paper by Slichter and Drickamer, the Gibbs energy was written in the form

$$G(T,P) = (1-\gamma)G_0(T,P) + \gamma G_1(T,P) - TS_{\text{mix}} + g_{\text{int}} + P\Delta V, \quad (2)$$

where $G_0(T, P)$ and $G_1(T, P)$ are the Gibbs potentials for the LS and HS phases, respectively, γ is the reduced amount of the HS phase, and S_{mix} is the configurational entropy

$$(S_{\text{mix}} = -k_B[\gamma \ln \gamma + (1-\gamma)\ln(1-\gamma)]),$$

P is the pressure, ΔV is the volume change, which is assumed to be equal $\gamma\Delta V_{HL}$ (ΔV_{HL} is the volume change upon the SC); g_{int} is the interaction energy. The interaction energy is calculated in the approximation of regular solutions

$$g_{\text{int}} = \Gamma\gamma(1-\gamma), \quad (3)$$

where Γ is the interaction parameter.

The phase equilibrium equation is obtained by equating to zero the derivative of the Gibbs energy with respect to the order parameter γ

$$dG(T,P)/d\gamma = -G_0(T,P) + G_1(T,P) + \Gamma(1-2\gamma) - k_BT\ln((1-\gamma)/\gamma) + P\Delta V_{HL} = 0, \quad (4)$$

or

$$\Delta H - T\Delta S + P\Delta V_{HL} + \Gamma(1-2\gamma) - k_BT\ln((1-\gamma)/\gamma) = 0, \quad (5)$$

where ΔH and ΔS are the changes in enthalpy and entropy upon the spin crossover at atmospheric pressure.

In Ref. 45, the pressure dependence of the fraction of the HS phase has been analyzed by expanding Γ to the second order in pressure. Generally, qualitatively correct dependences for the relative change of $\gamma(P)$ have been obtained, however no quantitative estimates of the parameters determining the pressure-induced SC have been carried out at the proper level due to the difficulties in solving analytically the obtained equations.

When the behavior of TISC under pressure is considered from the thermodynamic perspective, the state equation in the form of Eq. (5) is commonly applied. This, however, leads to at least two ambiguities. The first is related to the fact that the crossover temperature increases under pressure and, consequently, the enthalpy change increases as well. On the other hand, it is more or less impossible to explicitly obtain the enthalpy change using this equation. Naturally, this affects the understanding of the physics of the process. The second uncertainty stems from the fact that the at $T_{1/2}$ $\gamma = 1/2$ and the term $\Gamma(1 - 2\gamma) = 0$, i.e., the interaction, according to Eq. (5), does not contribute to a change in the crossover temperature under pressure. According to these considerations, the Clausius-Clapeyron equation can be applied to describe changes in the transition temperature under pressure

$$\Delta T_{1/2}/\Delta P = \Delta V_{HL}/\Delta S. \quad (6)$$

As follows from the experiments described above, the transition temperature is not always increased linearly with increasing pressure, and in some cases it is even decreased. If we exclude the experiments in which the $T_{1/2}(P)$ decreases with increasing pressure, then when the condition (6) is satisfied, there must be a correspondence between the SC pressure at room temperature ($P_{1/2}$) and the transition temperature at atmospheric pressure ($T_{1/2}$), specifically

$$P_{1/2} = \left[T_r - T_{1/2}\right] \cdot \left[\frac{\Delta S_{HL}}{(\Delta V_{HL})_P}\right]. \quad (7)$$

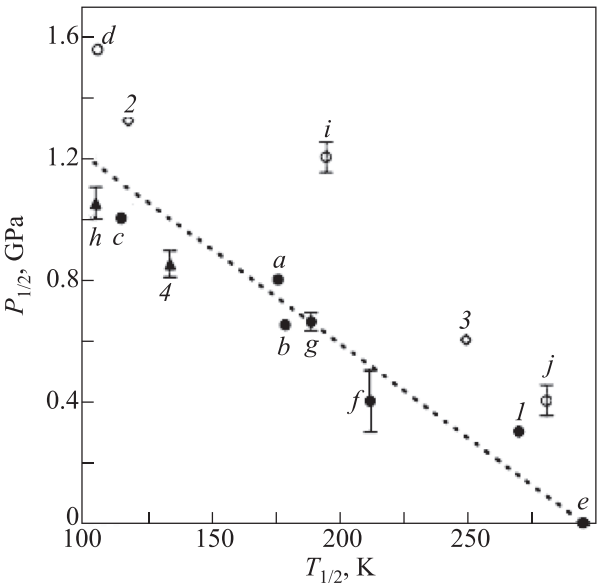


FIG. 16. Dependence of the spin transition pressure at room temperature ($P_{1/2}$) on the transition temperature at atmospheric pressure ($T_{1/2}$) for Fe (II)-containing complexes. There are four groups of compounds: with octahedral symmetry of the ligand environment (●); with distorted octahedra of the ligand environment (○); with structural phase transitions (◇), and polymeric compounds (▲): $Fe(Me_2bpy)_2(NCS)_2$ (1), $[Fe(2\text{-}pic)_3]Cl_2{\cdot}EtOH$ (2), $[Fe(2\text{-}pic)_3]Cl_2{\cdot}H_2O$ (3), $Fe(phen)_2(NCS)_2$ polymorph I (*a*) $Fe(phen)_2(NCS)_2$ polymorph II (*b*), $Fe(py)_2bpym(NCS)_2$ (*c*), $Fe(py)_2phen(NCS)_2$ (*d*), $Fe(bpy)_2(NCS)_2$ polymorph II (*f*), $Fe(PM\text{-}AzA)_2(NCS)_2$ (*g*), Fe (hyetrz)$_3$(3-nitrophenylsulfonate)$_2$ (h), $Fe[HB\{3,5\text{-}(CH_3)_2pz\}_3]_2$ (*i*), $[Fe(phy)_2](BF)_4)_2$ (*j*). *e* - point ($P = P_{atm}$, $T = T_{amb}$). Curve 4 represents a hypothetical behavior of $[Fe(btr)_2(NCS)_2]{\cdot}H_2O$ in the absence of a pressure-induced structural phase transition.

In the paper by Boillot *et al.*,[42] the applicability of this expression for a number of compounds has been analyzed. The results are shown in Fig. 16 as the dependence of $P_{1/2}$ on $T_{1/2}$ at atmospheric pressure. It can be seen, that a more or less linear behavior of $P_{1/2}(T_{1/2})$ is observed in mononuclear compounds with a presumably undistorted octahedron. The behavior of $P_{1/2}(T_{1/2})$ inconsistent with Eq. (7), which is observed for other compounds, has been explained by the presence of distortions or structural changes.

To eliminate these ambiguities, Spiering *et al.* have applied the mean-field model, where the spin-crossover molecules are considered placed in the field of elastic stresses. In this case the spin transition of one molecule alters the elastic energy by $\pm\Delta_{elast}$, and the interaction occurs only between the HS molecules.[38] Then the interaction term, which also takes into account the interaction with the field of elastic stresses, is

$$g_{int} = \Delta_{elast}\gamma - \Gamma\gamma^2. \tag{8}$$

Substituting the value of the interaction term in the form (8) into Eq. (2), the Gibbs free energy can be obtained as

$$G(T,P) = (1-\gamma)G_0(T,P) + \gamma G_1(T,P) - TS_{mix} + \Delta_{elast}\gamma - \Gamma\gamma^2 + P\Delta V. \tag{9}$$

Further, in the so-called Mainz model of cooperative elastic interactions,[13,36,46,85,86] the crystal is viewed as a homogeneous elastic medium with randomly distributed defects—HS and LS complexes, creating local stresses and causing isotropic deformation. Elastic medium is characterized by two constants: the bulk elastic modulus K and the Eshelby constant γ_0. The boundary of the sphere on which the stresses of the media and the defect are compensated determines the unstrained volume v_0. Position of the defects can be occupied by molecules in the HS and LS states with volumes v_{HS} and v_{LS}.

Taking into account the first terms of the expansion of the elastic energy proportional to the elastic modulus K, the authors have obtained the parameters Δ_{elast} and Γ explicitly[13,85]

$$\Delta_{elast} = \frac{1}{2}K\frac{(\gamma_0-1)}{\gamma_0}\frac{\delta v_{HL}}{v_c}(v_{HS}+v_{LS}-2v_0), \tag{10}$$

$$\Gamma = \frac{1}{2}K\frac{(\gamma_0-1)}{\gamma_0}\frac{\delta v_{HL}^2}{v_c}, \tag{11}$$

where $\delta v_{HL} = \gamma_0(v_{HS} - v_{LS})$ and v_c is the unit cell volume per molecule.

Considering only the terms proportional to γ, the Gibbs energy per single molecule is written as follows:[13]

$$g(\gamma,T) = (f_{HS} - f_{LS} + \Delta)\gamma - TS_{mix} - \Gamma\gamma^2 + P\delta v_{HL}\gamma, \tag{12}$$

where f_{HS} and f_{LS} are the free energy of a single molecule in the HS and LS states, respectively.

Introducing again the critical temperature $T_c = \Gamma/2k_B$, in Ref. 13 the expressions for $dT_{1/2}/dP$ and $d\sqrt{\Delta T_{1/2}}/dp$ have been obtained

$$\frac{dT_{1/2}}{dP} = (2\gamma^G H)\frac{T_c}{K}\left[\frac{1}{\sigma_c}A + \frac{\Delta C_{HL}}{2k_B\sigma_c}\right], \tag{13}$$

$$\frac{d\sqrt{\Delta T_{1/2}}}{dp} = (2\gamma^G H)\frac{\sqrt{T_c}}{K}\left\{T_c A + \frac{1}{2}\frac{\sqrt{\Delta T_{1/2}}}{T_c}\right\}. \tag{14}$$

Here $\sigma_c = \Delta S_{HL}/2k_B$, γ^G is the Grüneisen constant ($\partial K/\partial P = 2\gamma^G$), ΔS_{HL} and ΔC_{HL} are the changes in entropy and heat capacity upon the spin crossover

$$A = \frac{3\gamma_0 - 1}{2\gamma_0\dfrac{K}{(2\gamma^G+1)}\dfrac{\partial v_{HL}}{2k_BT_c} + \dfrac{\Delta_{elast} - 2k_B(T_c + \sigma_c T_{1/2})}{2k_BT_c}}. \tag{15}$$

After neglecting the quantities which are too small in magnitude, the dependences of two functions, α and β, which are proportional to the changes in the crossover temperature and the square root of the pressure hysteresis

$$\alpha = \frac{1}{T_c}\frac{K}{2\gamma^G+1}\frac{dT_{1/2}}{dP}, \tag{16}$$

$$\beta = \frac{1}{\sqrt{T_c}}\frac{K}{2\gamma^G H}\frac{d\sqrt{\Delta T_{1/2}}}{dP}, \tag{17}$$

on the parameter A, which are shown in Fig. 17. The intersection of α and β lines with the horizontal axis divides the plane into three regions. In the first region where $\alpha < 0$ and $\beta > 0$, the hysteresis increases and the crossover temperature decreases with increasing pressure. In the second region where $\alpha > 0$ and $\beta > 0$ both increase in the crossover temperature and hysteresis is possible, while in the third region

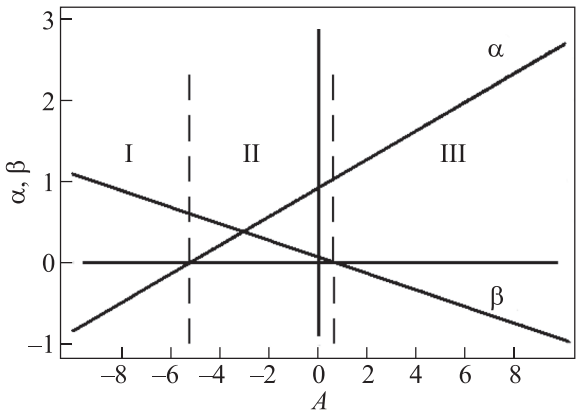


FIG. 17. Dependence of the dimensionless quantities α and β on the dimensionless quantity A. The vertical dashed lines separate three areas (I, II, III) with the different pressure dependences of the crossover temperature and hysteresis.[13]

where $\alpha > 0$ and $\beta < 0$, the crossover temperature increases and hysteresis decreases.

The first particularity of this approach is that it demonstrates different scenarios of possible behavior of $T_{1/2}$ and $\Delta T_{1/2}$ under pressure. The second particularity is that if the crossover temperature decreases with increasing pressure, the hysteresis necessarily increases (region I), and if the hysteresis decreases, the crossover temperature necessarily increases (domain III). On the other hand, there is a region (II), where both the crossover temperature and the hysteresis may increase.

Such conclusions from the theoretical consideration, which are absolutely contradictory to those available prior this study, fully reflect the experimentally observed behavior of TISC under pressure. The increase of the crossover temperature in Figs. 4 and 6 is accompanied by a reduction of the hysteresis, which corresponds to the region III in Fig. 17; the decrease of the crossover temperature in Fig. 5 is accompanied by an increase in the hysteresis (region I in Fig. 17). An increase in the hysteresis in the transition region in Figs. 4 and 6 is also accompanied by a decrease in the crossover temperature. At the same time, an increase in the hysteresis may also be accompanied by an increase of the crossover temperature as observed in Fig. 7 for $Fe(phy)_2(BF_4)_2$ (region II in Fig. 17).

Quantitatively, this model has been brought to test in Ref. 49 on $Fe(phen)_2(NCS)_2$, for which the most complete set of experimental parameters was available. Thus, all the parameters of the model could be expressed in terms of experimentally obtained data. It turned out that Eq. (11) for the parameter Γ does not allow to reproduce the experimentally observed values and does not correspond to the physical notions that Γ in the thermodynamic description in the case of hysteresis should be greater than $2k_BT_{1/2}$. This indicates that the expression for T and perhaps also that for Δ_{elast} need to be revised, although both expressions allow to obtain qualitatively correct results on the behavior of the crossover temperature and pressure hysteresis. If we assume that the expressions for Δ_{elast} and Γ correctly represent the true values of these parameters in the model of elastic interactions, then the obtained value of Γ is too small compared to the value of the interaction energy, required for the observed transitions. Therefore, we must assume that in addition to the elastic interactions, other interactions play an important role in TISC, which will be discussed in the microscopic consideration of SC.

To obtain the correct dependence of the crossover temperature on pressure and understand the physical nature of this change, we can apply a simple transformation to Eq. (5), which has been suggested in Ref. 87. Indeed, by adding and subtracting the term $\gamma\Gamma(T,P)$ in Eq. (5), the equilibrium equation can be obtained as

$$\Delta H(T,P) - T\Delta S + P\Delta V_{HL} + \Delta_{elast}(T,P) - \Gamma(T,P) + (1-2\gamma)\Gamma(T,P) + k_BT\ln((1-\gamma)/\gamma) = 0, \quad (18)$$

from which we can derive an expression for the pressure dependence of the crossover temperature

$$dT_{1/2}/dP = (\Delta V_{HL} + d(\Delta_{elast} - \Gamma)/dP)/\Delta S. \quad (19)$$

Equation (18), similar to Eq. (5), contains a term responsible for hysteresis $(1-2\gamma)\Gamma(T,P)$, which does not contribute to $T_{1/2}(P)$. However, it also contains the energy difference $(\Delta_{elast}(T,P) - G(T,P))$, which determines the behavior of $T_{1/2}$ on pressure and depends not only on Δ_{elast}, as was suggested earlier, but also on Γ. Equation (19) determines this influence through the change of Δ_{elast} and Γ under pressure. Moreover, if the change in Γ is positive, hence increasing the hysteresis, and large enough, the term $d(\Delta_{elast} - \Gamma)/dP$ may become negative, so the change in the crossover temperature with pressure will be negative as well, i.e. the crossover temperature may decrease with increasing pressure. This is possible only with a significant increase in the hysteresis.

An increase in the hysteresis may not lead to a decrease in the crossover temperature, but be accompanied by a reduction of the rate of the crossover temperature increase. Fig. 18 shows the temperature dependences $\chi_m T$ at different pressures for two compounds, $\{Fe[H_2B(pz)_2]_2(bipy)\}$ and $\{Fe[H_2B(pz)_2]_2(phen)\}$.[20] The former compound exhibits a hysteresis-free crossover, the temperature of which increases with increasing pressure. The latter compound has a crossover with a small hysteresis which initially increases with increasing pressure, and then decreases.

Fig. 19 shows the pressure dependences of the crossover temperatures for these compounds. It can be seen that while the first compound exhibits a linear dependence $T_{1/2}(P)$, for the second compound, the rate of increase of $T_{1/2}(P)$ first sharply drops with increasing the hysteresis, but with the hysteresis decreasing further, the rate of increasing of $T_{1/2}(P)$ increases again. This figure shows the direct impact of the interaction energy on the transition temperature.

Thus, all of the experimentally observed pressure dependences of the crossover temperature and hysteresis, regardless of the structural changes, can be described in the thermodynamic approximation.

To obtain more accurate values of the parameters Γ and Δ_{elast}, it is proposed to use an expression that determines the dependence of γ on the crossover temperature and is obtained directly from the equation of state (5)[73]

$$T = (\Delta H + P\Delta V_{HL} + \Delta_{elast} - 2\gamma\Gamma) \Big/ \left[\Delta S + k_B\ln\left(\frac{1-\gamma}{\gamma}\right)\right]. \quad (20)$$

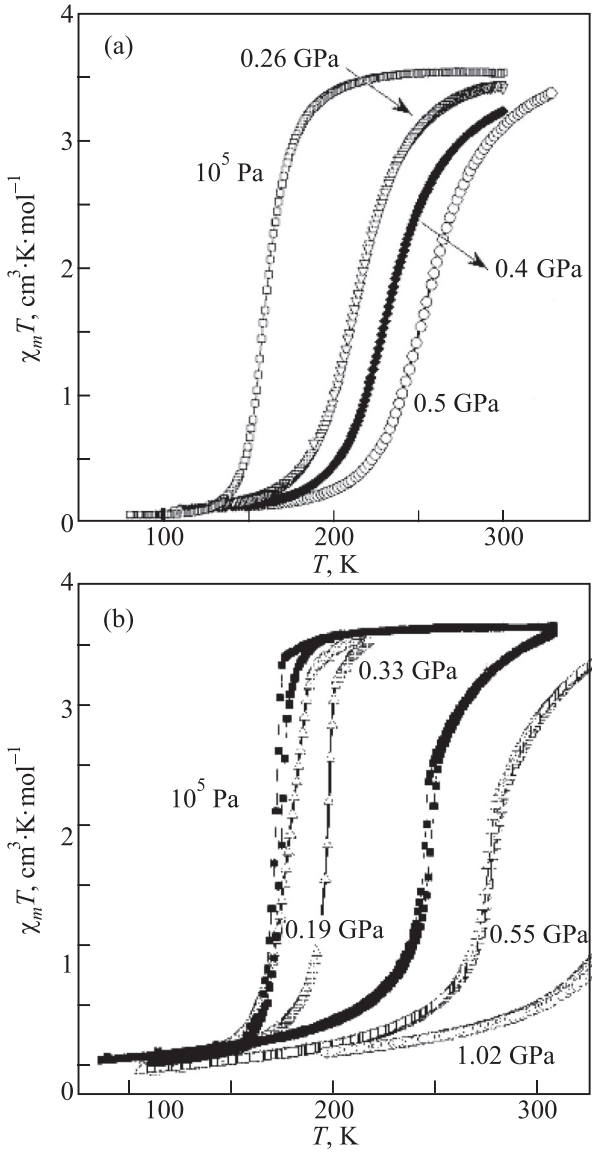


FIG. 18. Temperature dependences of $\chi_m T$ for {Fe[$H_2B(pz)_2]_2$(bipy)} (a) and {Fe[$H_2B(pz)_2]_2$(phen)} (b).[20]

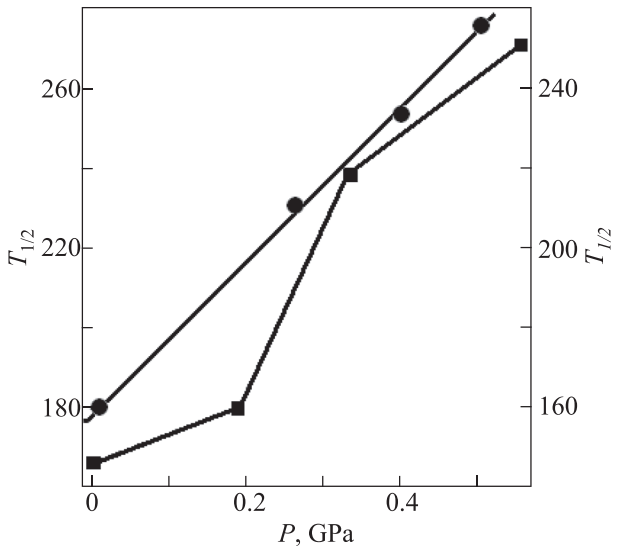


FIG. 19. Pressure dependence of $T_{1/2}$ for {Fe[$H_2B(pz)_2]_2$(bipy)} (●) and {Fe[$H_2B(pz)_2]_2$(phen)} (■).

To do this, Eq. (20) is fitted to the experimental curve $\gamma(T)$ at different pressures by adjusting the parameters Δ_{elast} and Γ. Using this approach, the dependences $\Delta_{\text{elast}}(P)$ and $\Gamma(P)$ in two isostructural compounds exhibiting 50% TISC, {Fe(pmd)H_2O[M$(CN)_2]_2$}·H_2O, where M = Ag or Au (Fe-Ag and Fe-Au). For these compounds, due to the 50% transition, expression (20) has the form

$$T = \frac{\Delta H + P\Delta V_{HL} + \Delta_{\text{elast}} + (1 + 0.5 - 2\gamma)\Gamma}{\Delta S + k_B \ln\big((1-\gamma)/(\gamma - 0.5)\big)}. \qquad (21)$$

In addition to these calculations, the dependences $T_{1/2}(P)$ and $\Delta T(P)$ for these two compounds (see Fig. 20) have also been obtained in Ref. 73. These graphs show a non-monotonic pressure dependence of $T_{1/2}$ and ΔT for both compounds.

Thus, the thermodynamic approach qualitatively correctly describes the behavior of TISC under pressure. An exception is the pressure-induced HS state in the region of the LS state, which has been observed in [Fe$(btr)_2(NCS)_2$]·H_2O (Fig. 8). The explanation for this behavior is only possible from the point of view of the microscopic theory, taking into account tetragonal and trigonal distortion of the octahedral environment of the central ion[67] and inelastic interactions of the spin-crossover molecules.[83]

Microscopic model of SC under pressure. The currently existing microscopic model allows to obtain the calculated dependences $\gamma(T)$ and $\gamma(P)$ for TISC and PISC, which correspond to the experimental data for specific spin-crossover systems.[69,82,83]

A molecular system in this model is regarded as a compound constructed of N complex molecules forming a cubic

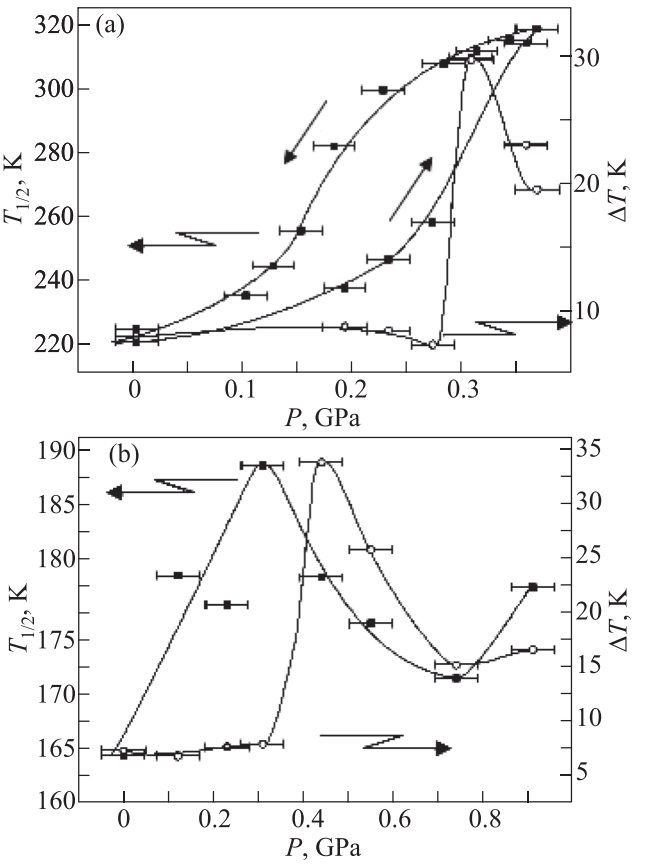


FIG. 20. Pressure dependences of the crossover temperature $T_{1/2}$ (■) and the width of hysteresis ΔT (○) for Fe-Ag (a) and Fe-Au (b).[73]

lattice and placed in an elastic medium. A complex molecule contains a spin-active ion in the octahedral environment of ligands. The model takes into account the effect of temperature, external pressure and parameters of the theory describing a fully symmetric strain of both ligand complexes and elastic medium on the probability density of the spin states of a 3*d*-ion. The distortions of complexes which are different from the fully symmetric are not considered in the model. This approximation is justified by the fact that the SC is primarily determined by the difference in the volume of ligand complexes in the HS and LS states regardless of the specific type of deviation from the octahedral symmetry of the spin-crossover molecules.

According to the model concept, at the microscopic level, which takes into account local deformations of the complexes and, in the continuum approximation, the elastic deformation of the medium, the HS and LS states are described by the corresponding eigenvalues of the reduced Hamiltonian, which is represented in the form

$$H = \sum_{i=1}^{N}\left[f_0(i) + V_0(i) + \left(\frac{\partial V(i)}{\partial Q_1}\right)_0 Q_1 + \left(\frac{\partial V(i)}{\partial U_1}\right)U_1\right] + \frac{1}{2}NM_1\omega_0^2Q_1^2 + \frac{1}{2}NC_1U_1^2. \quad (22)$$

Here $f_0(i)$ is the Hamiltonian of a free metal ion in the i-th complex; $V_0(i)$ is the potential of a ligand complex in the absence of deformation; $V(i)$ is the ligand field at the i-th node; Q_1 is the fully symmetric deformation of the complex; M_1 is the effective mass for the normal vibration of ligands with the eigenfrequency ω_0; U_1 is the elastic stress in the medium, $C_1 = 3BV_0$ (B is the bulk elastic modulus, V_0 is the average volume of the spin-crossover complex in the absence of deformation).

To circumvent the fundamental difficulties in working with the density matrix, in the model, the eigenfunctions and eigenvalues of the Hamiltonian, which are defined on the basis of the LS $\Rightarrow$ $|{}^1A_{1g}\rangle \equiv |\psi_l\rangle$ and HS $\Rightarrow$ $|{}^5T_{2g}\rangle \equiv |\psi_h\rangle$, are expressed in terms of quantum and thermodynamic averages of the variables responsible for the deformation of two types: local deformation of the spin-active complex and lattice strain.

"Inelastic forces" q_1 are determined by the eigenvalues of the operators of interaction of d-electrons of the metal ion with the ligand field through the interaction operator $\partial V(i)/\partial Q_1$. The variable q_1 takes into account, besides the intramolecular interaction, also the indirect effect on the d-electron deformation of the crystal field of neighboring complexes. "Elastic forces" u_1 describe the interaction of d-electrons with an elastic medium and are determined by the interaction operator $\partial V(i)/\partial U_1$. Both variables are expressed through quantum-mechanical averages—eigenvalues of the interaction operators, and through the thermodynamic averages, which are defined by the occupation of energy levels, which are obtained from the condition of thermodynamic equilibrium—minimum of the Gibbs potential. In turn, the condition of thermodynamic equilibrium self-consistently includes the eigenvalues of the energies of the respective spin states.

Since the volumes of complexes in the HS and LS states are different, then the change of the spin state of a molecule alters the elastic stress of the medium and, thus, u_1, in the same way as q_1, is responsible for the cooperative interaction of complex molecules. The most likely variables of the q_1 and u_1 theory are the quantities satisfying the conditions of thermodynamic equilibrium $\partial G/\partial q_1 = 0$ and $\partial G/\partial u_1 = 0$, where G is the Gibbs free energy

$$u_1 = -p - b(\rho_L + \lambda\rho_H) = -p - b[1 + (\lambda - 1)\rho_H], \quad (23)$$

$$q_1 = -b_1[\rho_L + \lambda_1\rho_H] = -b_1[1 + (\lambda_1 - 1)\rho_H]. \quad (24)$$

The occupation of the high-spin state is the quantity

$$\rho_H = \frac{1}{1 + \frac{g_L}{g_H}e^{\Delta E_{HL}/kT}}, \quad (25)$$

when the following normalization condition is satisfied:

$$\rho_L + \rho_H = 1,$$

electronic degeneracy is denoted as g_L and g_H, respectively, the parameter p is the reduced pressure $p = \sqrt{3}PV_0/\sqrt{C_1}$ (it is assumed that C_1 is independent of P); model parameters b, b_1, λ, λ_1 are responsible for inter- and intra-complex interaction of molecules in the HS and LS states.

In Ref. 83, the relation of the energy g_{int}, written in the form (8), with the phenomenological model Eq. (2) has been established and the following relations between macro- and microscopic parameters have been found:

$$\Delta V_{HL} = -\frac{\sqrt{3}}{\sqrt{C_1}}V_0 b(\lambda - 1), \quad (26)$$

$$\Delta_{\text{elast}} = -\frac{1}{2}\omega_0 b_1^2(\lambda_1^2 - 1) - b^2(\lambda - 1) - b_1^2(\lambda_1 - 1), \quad (27)$$

$$\Gamma = \frac{1}{2}\left[b^2(\lambda - 1)^2 + b_1^2(\lambda_1 - 1)^2\right]. \quad (28)$$

Note that both Eqs. (27) and (28) include the elastic (determined by the parameter b) and inelastic (which enters the equations through the parameter b_1) interactions, as well as the vibrational energy of complexes in the long-wavelength limit (which enters through the parameter ω_0). So here Δ_{elast} is responsible for the field in the lattice, which includes both elastic and inelastic parts, and, therefore, should be renamed $\Delta_{\text{elast}} \to \Delta_{\text{int}}$. The macroscopic parameter Γ is now responsible not only for the elastic but also inelastic cooperative interaction of spin-crossover complexes.

In the paper, the dependences of the macroscopic parameters on the experimentally measured quantities have been obtained

$$\Gamma(T_{1/2}, P) = ak_BT_{1/2}(P), \quad (29)$$

where

$$a = 2[1 - (\Delta S)^2/[4k_BC_P(T_{1/2})]], \quad (30)$$

$$\Delta_{\text{int}} = \left(\frac{\Delta S + ak_B}{ak_B}\right)\Gamma - P\Delta V_{HL} - \Delta\varepsilon_0. \quad (31)$$

The theory has been evaluated by comparing the calculated curves $\rho_H(T, P)$ with similar curves based on the

experimentally determined temperature dependences of χT at different pressures P.

By solving the equation

$$\rho_H = \frac{\rho_0}{1 + r_{\text{eff}} \exp\left(\frac{-\Delta E_{HL}}{kT}\right)} + \frac{1}{1 + \frac{1}{r_{\text{eff}}} \exp\left(\frac{\Delta E_{HL}}{kT}\right)}, \quad (32)$$

with respect to ρ_H, the dependence $\rho_H(T)$ for a given P can be obtained for TISC, or the dependence $\rho_H(P)$ for a given T can be obtained for PISC. Equation (32) takes into account the cases of incomplete SC, when in the entire experimental range of T and P there is always a certain fraction of HS complexes, which corresponds to ρ_0 in Eq. (32). Furthermore, in the case of PISC, when ΔE_{HL} is calculated, in Eq. (23) P will vary, and in Eq. (31) $P = P_{1/2}$. The pre-exponential factor r_{eff} is a fitting parameter. In a first approximation

$$r_{\text{eff}} = \left(\frac{z_H^{\text{vib}}}{z_L^{\text{vib}}} \cdot \frac{g_2}{g_1}\right),$$

where z_H^{vib} and z_L^{vib} are the partition function of the vibrational states of molecules in the HS and LS phases, respectively.

The calculations have been performed for a number of compounds, and the behavior of both TISC and PISC have been investigated: molecular crystal $Fe(phen)_2(NCS)_2$,[23,48] 2D coordination polymers.[70,80] In the case of TISC the data on magnetic susceptibility have been used and the behavior of PISC has been experimentally investigated by absorption spectroscopy at 297 K. The data of calorimetric measurements for $[Fe(phen)_2(NCS)_2]$ were taken from Ref. 88, and the data for 2D-polymer were taken form Ref. 70.

As shown experimentally for these 2D polymer systems, the positions of the points of equal concentrations on the SC curves, which are obtained by different ways of inducing the transition on one and the same material might not match. For instance, in the case of TISC in Pd3Fpy, $T_{1/2} = 300.3$ K at $P = 0.12 \pm 0.05$ GPa, while in the case the PISC, $P_{1/2} = 0.33 \pm 0.05$ GPa at $T = 297$ K. Similarly for Pt3Fpy: $T_{1/2} = 332.9$ K at $P = 0.3$ GPa (TISC), $P_{1/2} = 0.3$ GPa at T = 297 K (PISC). The points of equal concentrations coincide only in one parameter (P or T), while mismatch of the second thermodynamic parameter is the result of a difference in the method of inducing the SC.

The calculations following the proposed microscopic model have shown that in all these compounds in the case of TISC, the inelastic forces $|q_1|$ are always greater than the elastic forces $|u_1|$. An applied pressure tends to level $|q_1|$ and $|u_1|$. However, in the case of PISC the pattern of action of elastic and inelastic forces changes. For the compound $Fe(phen)_2(NCS)_2$, in which the positions of points of equal concentration on the P–T-diagram are nearly independent on the method of induction, the values of $|q_1|$ and $|u_1|$ are of the same order in the whole range of temperatures and pressures. In 2D polymers, the magnitude of elastic interaction in the case of PISC turns out to be much greater than $|q_1|$: in Pd3Fpy, which exhibits the most pronounced difference in the phase behavior depending on the method of induction, the values of $|u_1|$ exceed the values of $|q_1|$ by two orders of magnitude, while for Pt3Fpy the values of $|u_1|$ exceed the values of $|q_1|$ by more than an order of magnitude.

From this we can conclude that the development of TISC is primarily determined by inelastic intra- and inter-complex interactions, while the behavior of PISC is mainly determined by the elastic energy of the medium and the elastic interactions between spin-crossover molecules.

In the last decade, a number of studies have appeared that employ the theory of electron density potential. Calculations of force-frequency characteristics of the vibrational modes in the first coordination sphere have been carried out for specific compounds. For $[Fe(PM\text{-}BIA)_2(NCS)_2]$, the characteristics of spin-crossover have been obtained and the regions of existence of different phases have been identified using the theory of electron density potential and molecular dynamics.[90] This is the first study in which the P–T-diagram of a spin-crossover compound has been calculated.

4. Conclusion

Drawing on the quote by one of the most prominent scientists in the field of spin-crossover phenomena Philipp Gütlich, "phase transition high spin–low spin is one of the most fascinating games of electrons in the coordination chemistry of 3*d*-elements," we can conclude after analyzing its manifestations under pressure that it is also one of the most wonderful manifestations of physical effects on solid. Indeed, such a variety of changes in phase transition under external influences cannot be observed when studying any other phenomenon. This review shows that the temperature, hysteresis, slope of the transition, symmetry, and other parameters of the HS–LS phase transition can be adjusted over a wide range by applying physical actions. This opens up new ways of obtaining molecular complex compounds for practical applications in modern electronics, which are based not only on the use of charge or spin of a free electron, but also on the spin change of a bound electron.

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